

EXACT AND APPROXIMATE NUMBERS

ACCURACY AND PRECISION, SIGNIFICANT DIGITS, AND ROUNDING

I. Exact and Approximate Numbers

Exact number

1. Any number **obtained by a counting process**. Example: 60 students in a class — 60 is an exact number
2. Any number **given by definition**. Example: One hour equals 60 minutes — 60 is an exact number

Approximate Number

1. Any number **obtained by a process of measurement**
Example: The distance between two cities is 60 miles — 60 is an approximate number.
The distance may be a little more or a little less than 60 miles.

II. Accuracy and Precision

Accuracy of an approximate number is **judged by the number of significant digits**.

Example: 4.63 cm is more accurate than 4.6 cm.

Precision of an approximate number **refers to the decimal position of the significant digit furthest to the right**

Example: 26.53 is more precise than 2512 (26.53 was measured to hundredths, 2512 to units)

III. Significant Digits are those digits that have meaning relative to the measuring process used to obtain them.

Five rules governing which digits of an approximate number are significant:

1. **Nonzero digits** are always significant
2. **Zeros that are preceded and followed by significant digits** are always significant.
3. **Final Zeros to the right of a decimal point** are significant
4. **Final zeros on a whole number** are not significant (unless further information indicates otherwise).
5. **If no digits left of the decimal point**, zeros between the decimal point and the first digit are not significant.

Examples

1. 14.24 has **four** significant digits (**Rule 1**)
2. .0036 has **two** significant digits (**Rule 1 & 5**)
3. 14.0024 has **six** significant digits (**Rule 3**)
4. 7001 has **four** significant digits (**Rule 2**)
5. 8.9000 has **five** significant digits (**Rule 3**)
6. 0.03600 has **four** significant digits (**Rule 1 & 5**)
7. In the statement, "Hawaii is 2400 miles west of Los Angeles," there are **two** significant figures (**Rule 4**).
8. If the distance to the nearest mile between cities is 20 miles, there are **two** significant figures (**Rule 4**).

IV. Rounding to Indicated Place Value

1. If the digit to the right of the indicated place value is less than 5 (0, 1, 2, 3, 4), retain the digit at place value.
2. If the digit to the right of the indicated place value is greater than or equal to 5 (5, 6, 7, 8, 9), add 1 to the digit.
3. Eliminate all digits to the right of the indicated place value.

Examples:

- | | |
|--|---|
| <ol style="list-style-type: none"> 1. Round 2.57834 to the nearest hundredth <ol style="list-style-type: none"> i. 7 is hundredth digit ii. 8 is first digit to the right iii. ANSWER: 2.58 (Rule 2) | <ol style="list-style-type: none"> 2. Round 37.50 to nearest unit <ol style="list-style-type: none"> i. 7 is unit digit ii. 5 is first digit to the right iii ANSWER: 38 (Rule 2) |
| <ol style="list-style-type: none"> 3. Round 0.0031 to the nearest thousandth <ol style="list-style-type: none"> i 3 is thousandth digit ii 1 is first digit to the right iii ANSWER: 0.003 (Rule 1) | |

(Over)

V. Rounding in Calculations**Adding or Subtracting approximate numbers:** Express result with the **precision of the least precise number**

Example: add 73.2, 8.0627, 93.57, 66.296
 73.2 is least precise (precise to tenths place)
 241.1286 is total
 ANSWER 241.1

Example: subtract 36.1 from 727.842
 36.1 is less precise (precise to tenths place)
 691.742 is result of subtraction
 ANSWER: 691.7

Multiplying or Dividing approximate numbers : Express result **to the accuracy of the least accurate number**

Example: divide 292.6 by 3.4
 3.4 is least accurate (only two significant digits)
 86.05824 is result of division
 ANSWER: 86 (two significant digits)

Example: multiply 2.4832 by 30.5
 30.5 is less accurate (only three significant digits)
 75.7315 is result of multiplication
 ANSWER: 75.7 (three significant digits)

Taking the root of an approximate number : Express result **to the accuracy of the number**

Example: take $\sqrt{3.7}$ (two significant digits)
 Result is 1.9235384
 ANSWER: 1.9 (two significant digits)

Rounding before and after calculations:

Before calculation, all numbers except the least precise or least accurate may be rounded to one place beyond that of the least precise or least accurate

Example:	rounding before calculation	
73.2	least precise = tenths place	73.2
8.0627	rounded to hundredths →	8.06
93.572	rounded to hundredths →	93.57
66.296	rounded to hundredths →	66.30
241.1287		241.13
	rounding after calculation	ANSWER: 241.1

Note: if calculator is used, rounding as shown above need not be followed.

Exact number in a calculation: If an exact number is included in a calculation, there is no limit to the number of decimal places it may take. The **accuracy of the result is limited only by the approximate numbers involved.**

Example: consider the **exact number 144** and the **approximate number 2.7**.

If these are **added or subtracted**, express result in tenths:

$$144 + 2.7 = 146.7 \qquad 144 - 2.7 = 141.3$$

If these are **multiplied or divided**, express result to two significant digits:

$$144 \times 2.7 = 390 \qquad 144 \div 2.7 = 53$$

(Over)